

Linear Regression Analysis for Medical Aid Charges

Introduction

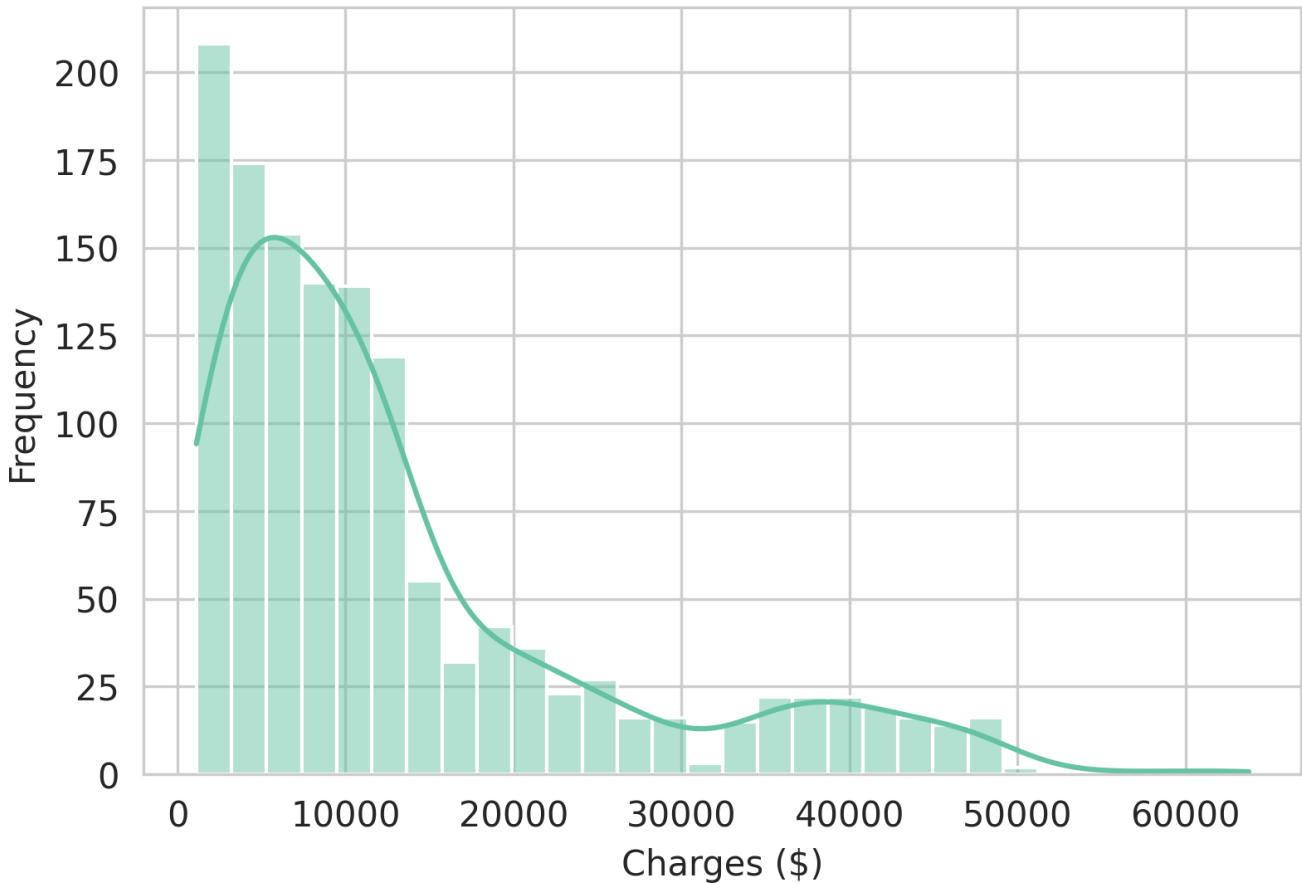
This report presents a linear regression model to predict medical aid charges, supporting the South African medical aid scheme’s goal of tailoring costs based on client lifestyles and regions. The Kaggle insurance dataset (1,338 records) includes features (age , sex , bmi , children , smoker , region) and the target variable (charges), serving as a proof-of-concept for data-driven pricing (The Independent Institute of Education, 2025). Developed in Python, the analysis identifies key cost drivers to inform strategic pricing decisions.

Exploratory Data Analysis

Exploratory data analysis (EDA) revealed critical patterns, visualized using the accessible “Set2” palette (Seaborn, 2021):

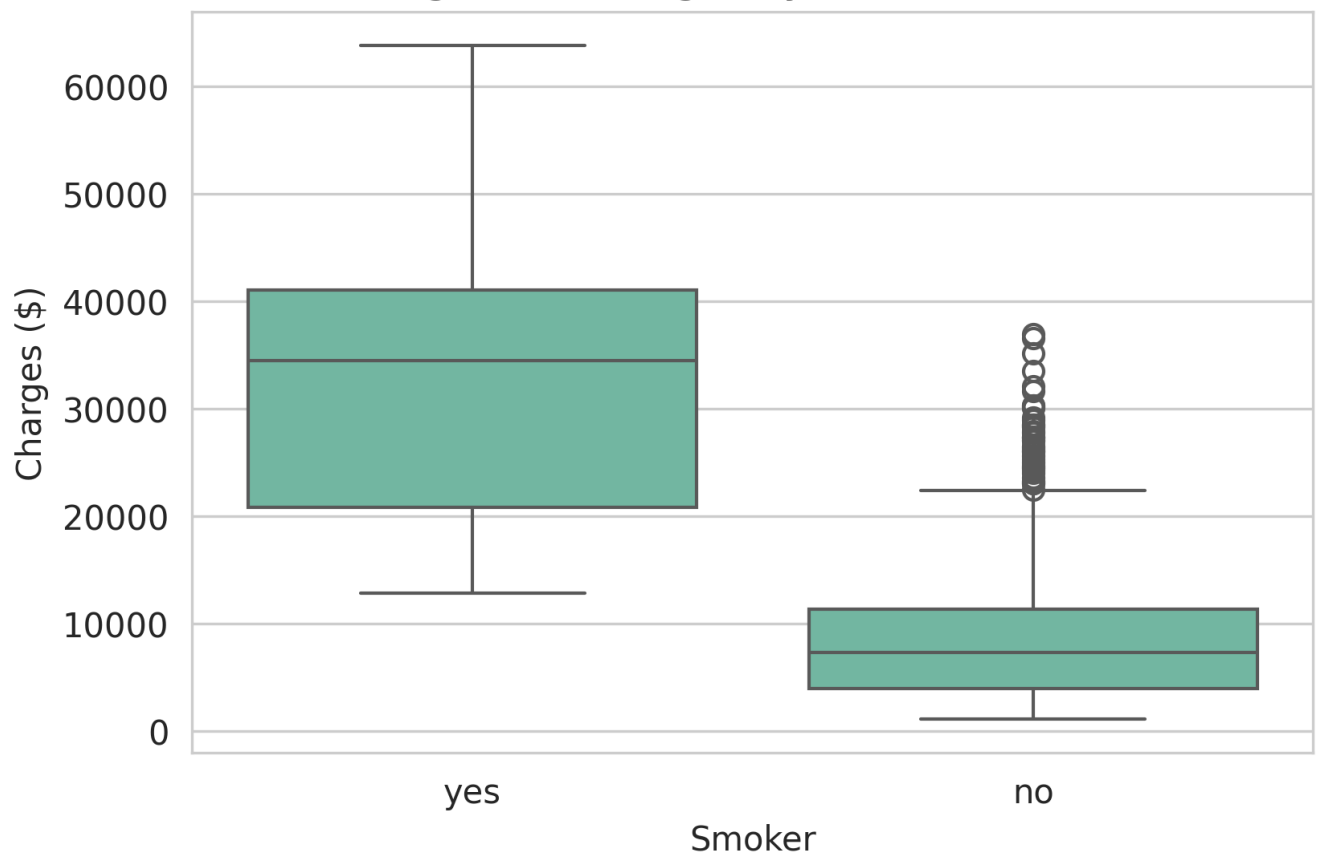
- Distributions:** Charges are right-skewed (mean: \$13,270, max: \$63,770), indicating high-cost outliers.

Figure 1: Distribution of Charges



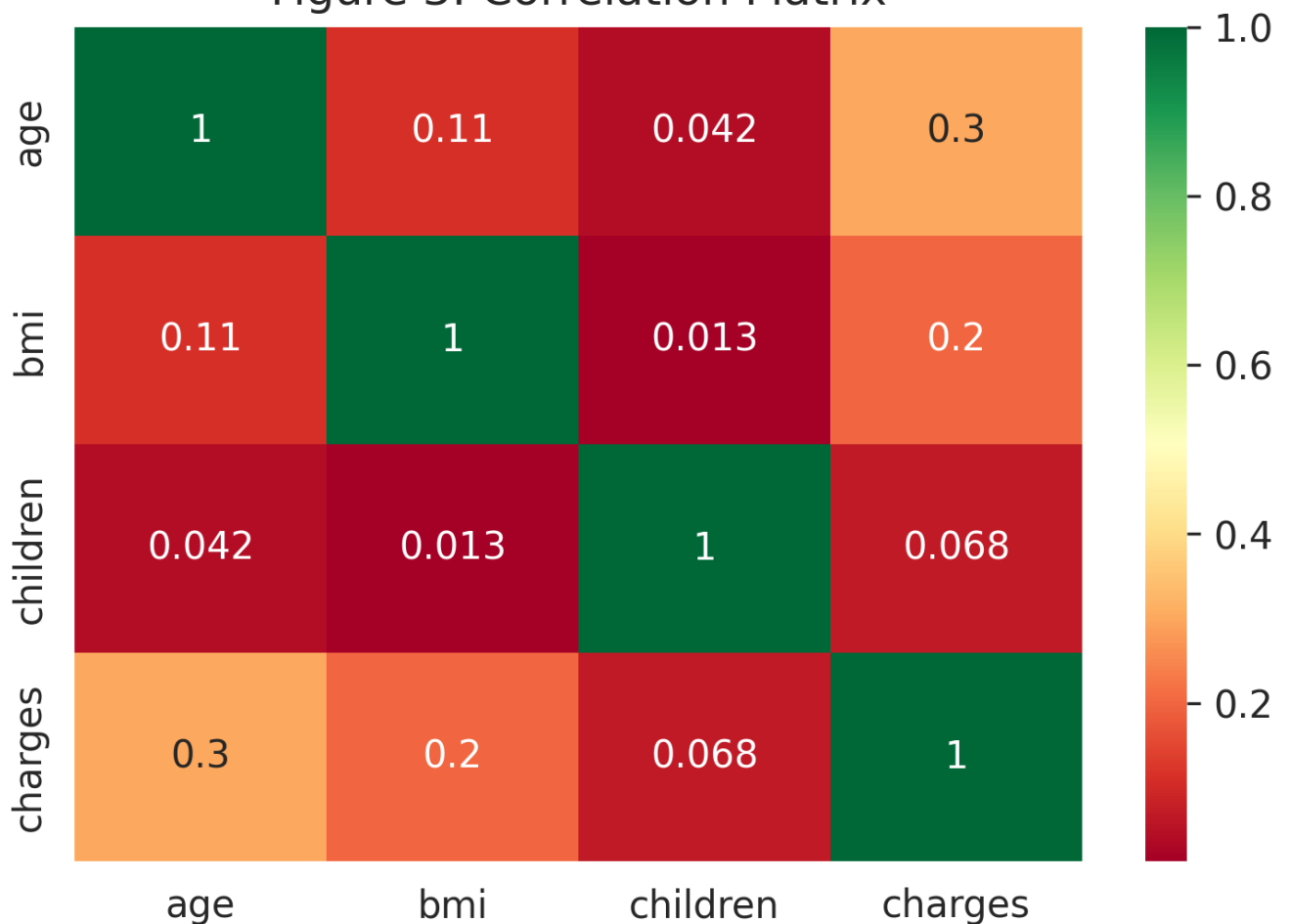
- Patterns:** Smokers incur significantly higher charges (median ~\$34,000 vs. ~\$7,000 for non-smokers), a key pricing insight.

Figure 2: Charges by Smoker Status



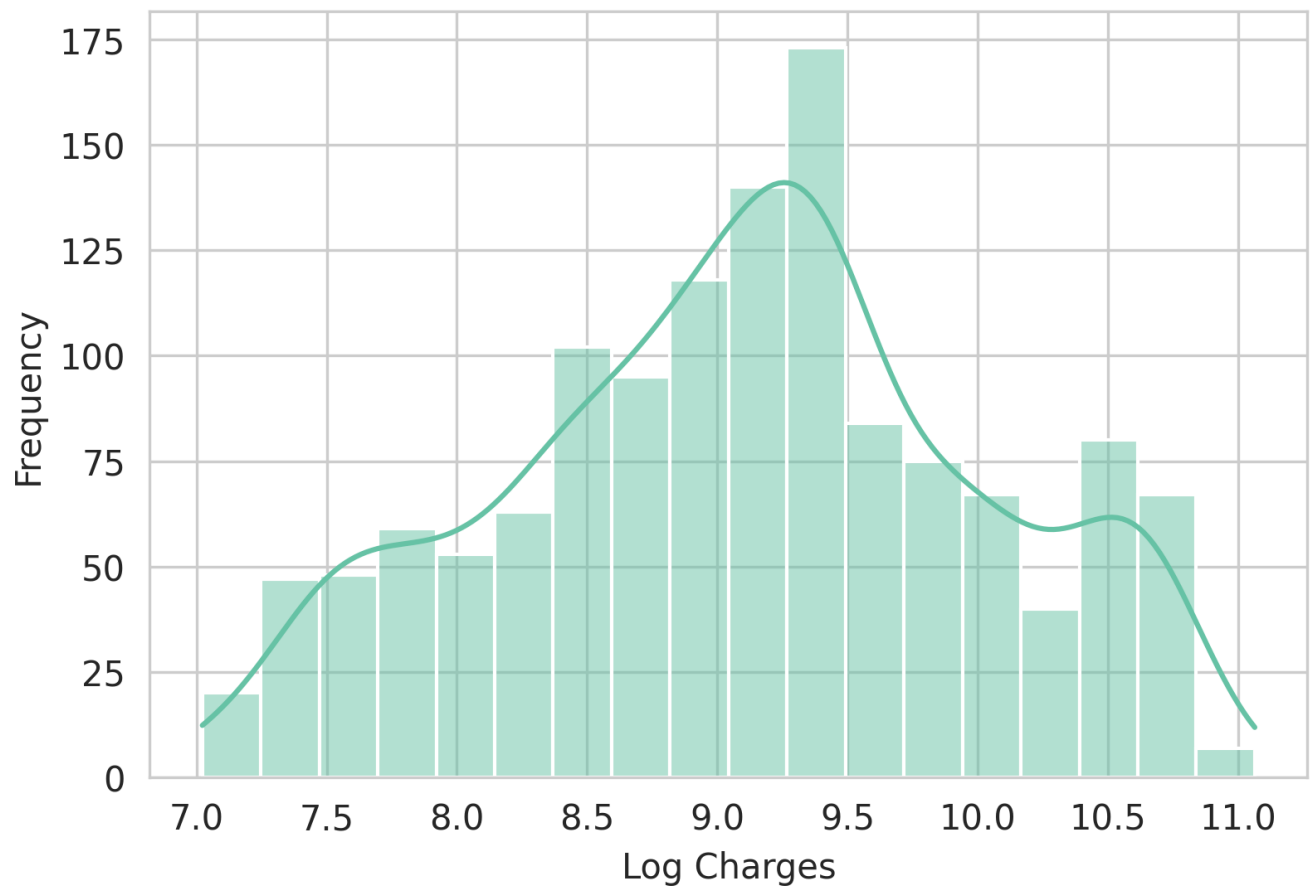
- **Correlations:** age (0.3) and bmi (0.2) moderately predict charges, while children has a weaker effect.

Figure 3: Correlation Matrix



- **Preprocessing:** Categorical variables (sex , smoker) were encoded using LabelEncoder, and region was one-hot encoded (Brownlee, 2020). Log-transformation normalized charges' distribution.

Figure 4: Distribution of Log-Transformed Charges



Smoking and age emerged as primary cost drivers, guiding feature selection.

Model Training

A linear regression model was trained using Scikit-Learn, with features selected via correlation analysis and p-values (<0.05) for statistical significance (David, 2021). An interaction term (smoker*BMI) captured the combined effect of surfing and BMI. The dataset was split (80% training, 20% test) to prevent overfitting. Performance metrics show a strong fit:

- Mean Squared Error (MSE): 20,919,716.75
- Root Mean Squared Error (RMSE): 4,573.81
- R²: 0.87 (explaining 87% of variance in charges)

Model Evaluation

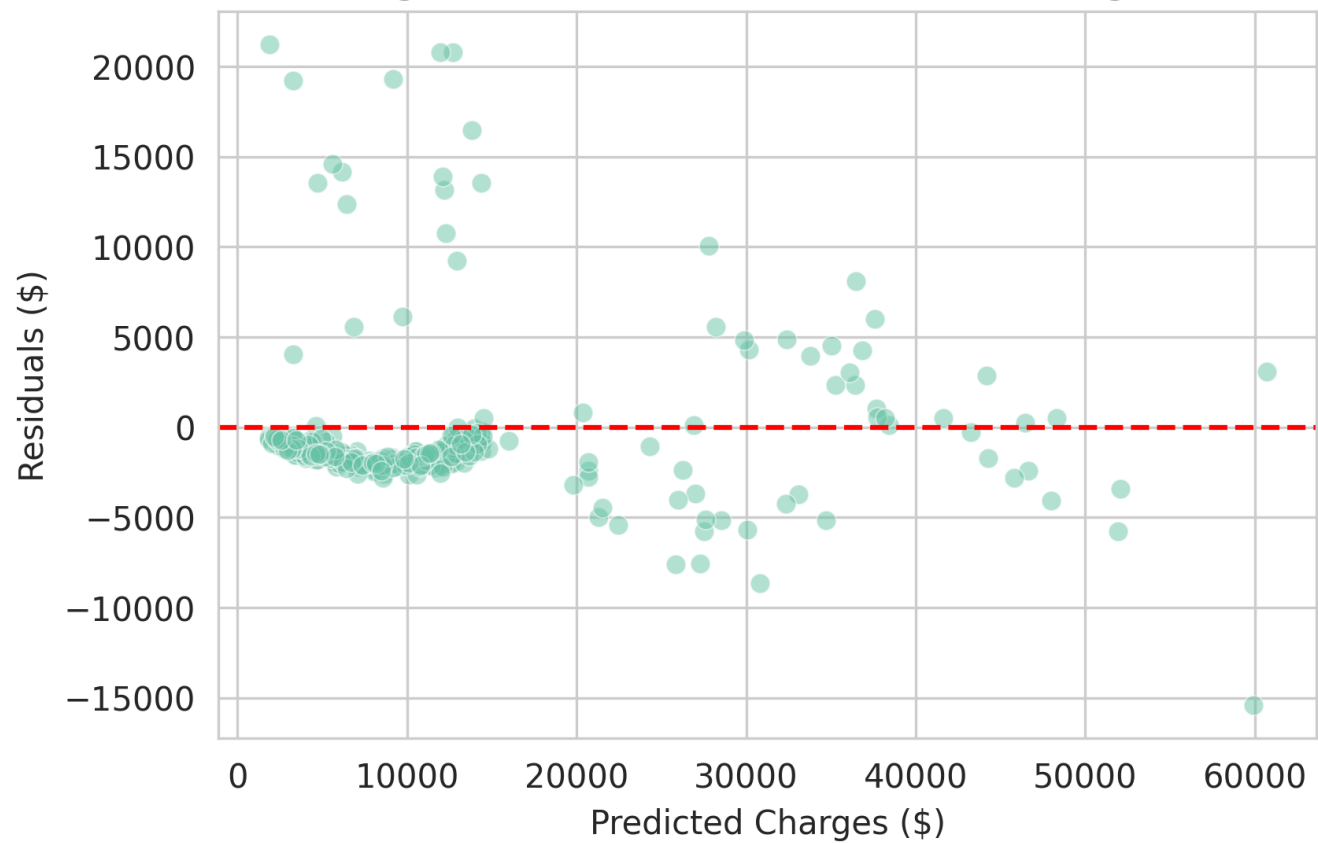
The model was evaluated for robustness (Srivastava, 2019):

- **Metrics:** R² (0.87) and Adjusted R² (0.86) confirm high accuracy across nine predictors.
- **Coefficients** (Table 1):

Feature	Coefficient
age	263.39
sex	-525.23
bmi	20.25
children	463.65
smoker	-21,206.91
region_northwest	-631.42
region_southeast	-967.48
region_southwest	-1,233.43
smoker_bmi	1,470.86

- **Interpretation:** The `smoker_bmi` coefficient (1,470.86) indicates that for smokers, each BMI unit increases charges by ~\$1,471, offset by the `smoker` coefficient (-21,206.91), yielding a net positive effect (e.g., ~\$44,125 for BMI=30). The `age` coefficient (263.39) shows a \$263 increase per year, critical for pricing.
- **Residuals:** Random scatter confirms model assumptions (e.g., homoscedasticity).

Figure 5: Residuals vs. Predicted Charges



- **Retrain:** Not required ($R^2 \geq 0.7$), though log-transformation was tested (Brownlee, 2020).

Conclusion and Recommendations

Smoking and age drive medical aid charges, with smokers incurring higher costs (median ~\$34,000). The model (R^2 : 0.87, Adjusted R^2 : 0.86) is reliable for pricing predictions. Recommendations include:

- Prioritize smoking status and age in pricing, adjusting premiums for high-BMI smokers.
- Integrate South African data to refine predictions, addressing US dataset limitations.
- Update the model regularly with new data for sustained accuracy.

This proof-of-concept enables data-driven pricing, enhancing client value and efficiency.

References

Brownlee, J., 2020. How to transform target variables for regression. [online] Available at: <https://machinelearningmastery.com/how-to-transform-target-variables-for-regression-in-python/> [Accessed 30 April 2025].

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